

IFN647 Week 8 Review Questions

Solutions

Q1 Which of the following descriptions is wrong? and justify your answer.

- (1) In the query likelihood retrieval model, documents are ranked by the probability that the query text could be generated by the document language model.
- (2) To use the unigram language model to estimate $P(Q|D)$ - the document D 's score for the given query $Q = \{q_1, q_2, \dots, q_n\}$, the language model probabilities $P(q_i|D)$ ($i=1, 2, \dots, n$) are needed to be estimated.
- (3) In contrast to the query likelihood model, the document likelihood model $P(D|Q)$ is more theoretical in the context of information filtering, because there is no explicit query available.
- (4) Kullback-Leibler (KL) divergence is a technique for avoiding this estimation problem and overcoming data sparsity, which means that we typically do not have large amounts of text to use for the language model probability estimates.

Solution: (4)

Kullback-Leibler (KL) divergence is a measure to estimate the difference between two probability distributions.

Smoothing is a technique for avoiding this estimation problem and overcoming data sparsity, which means that we typically do not have large amounts of text to use for the language model probability estimates.

Q2 (Pseudo relevance feedback)

Assume your user provides a query $Q = \{q_1, q_2, \dots, q_m\}$ and a data collection $C = \{d_1, d_2, \dots, d_N\}$ (unlabelled dataset) and asks you to define a probabilistic IF model by using pseudo relevance feedback techniques. After your data exploration of C , and you find that basically the distribution of terms in relevant documents is independent and their distribution in non-relevant documents is independent. And you want to define the probable relevance based only on the presence of search terms in documents. Describe your thoughts on how to design this IF model.

Tip: Review the Pseudo-Feedback Algorithm in a week 5 lecture notes "IR models".

Solution:

Phase 1: Generate a training set D using the Pseudo-Feedback Algorithm

By using pseudo relevance feedback, we use the following process to generate a training set D :

- (1) rank documents in C in descending order by using the query Q (you can choose an IR model to do it).
- (2) select top- k_1 documents (e.g., $k_1 = 20$) as Rel_1 , a set of possible relevant documents.
- (3) select important topical features T from Rel_1 (e.g., words with high $tf \cdot idf$ weight)
- (4) For each word w in T

$$P(w|R) \propto P(w, q_1, \dots, q_n) = \frac{\sum_{d \in Rel} P(d)P(w|d) \prod_{i=1}^m P(q_i|d)}{\sum_{d \in Rel} P(w|d) \prod_{i=1}^m P(q_i|d)}$$

- (5) Rank documents d in Rel_1 using the KL-divergence score

$$\sum_{w \in d} P(w|R) \log P(w|d)$$

- (6) select top- k_2 (e.g., $k_2 = 10$) documents as Rel_2 .

- (7) Let $D = C$, $D^+ = Rel_2$ and $D^- = C - Rel_2$.

In the above process, $P(d)$ is usually assumed to be uniform and can be ignored, $P(w|d) = tf(w, d)$, and $P(q_i|d) = tf(q_i, d)$. You may use smoothing technique to refine $P(w|d)$ and $P(q_i|d)$. For example,

$$P(w|d) = (1 - \lambda)tf(w, d) + \lambda tf(w, C)$$

Phase 2: Design a probabilistic IF model (review week 7 lecture notes)

Based on **I2** (the distribution of terms in relevant documents is independent and their distribution in non-relevant documents is independent) and

O1 (the probable relevance is based only on the presence of search terms in documents)

We can choose the following term weighing formula

$$w_2(t) = \log \frac{\frac{r(t)}{R}}{\frac{n(t) - r(t)}{N - R}}$$

where $R = |D^+|$, or $R = \text{len}(D^+)$, $N = |D|$, or $N = \text{len}(D)$,
 $n(t) = |\{d \in D, t \in d\}|$ or $n(t) = \text{len}([d \text{ for } d \text{ in } D \text{ if } t \text{ in } d])$, and
 $r(t) = |\{d \in D^+, t \in d\}|$ or $r(t) = \text{len}([d \text{ for } d \text{ in } D^+ \text{ if } t \text{ in } d])$.

Q3 (Pseudo-Relevance Feedback Style CoT Prompt)

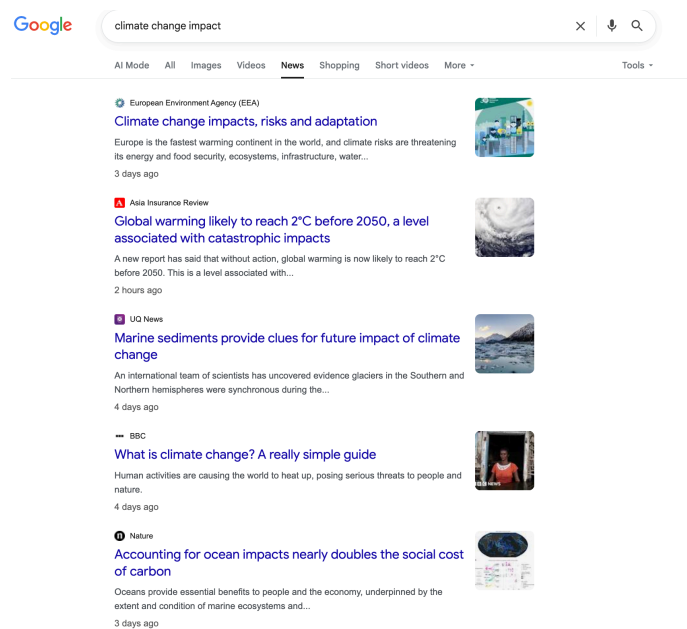
The following is a prompting template for query expansion using pseudo-relevance feedback.

Task: Expand the query using pseudo-relevance feedback.
Query: [Original query]
Top documents: [Short snippets]
Instructions: Briefly explain your reasoning in **2-3 steps**, then provide the expanded query.

Design a Pseudo-Relevance Feedback-style Chain-of-Thought (CoT) prompt for the original query “climate change impact”, and test it in ChatGPT.

Tip: You may use Google News search to collect a few short snippets. Use these snippets to compare the search results obtained with the **original query** versus the **expanded query**.

Solution:



Top-5 Snippets from Google news search:

- “Climate change impacts, risks and adaptation”,
- “Global warming likely to reach 2°C before 2050, a level associated with catastrophic impacts”,
- “Marine sediments provide clues for future impact of climate change”,
- “What is climate change? A really simple guide”,
- “Accounting for ocean impacts nearly doubles the social cost of carbon”

Prompt: Assume the top retrieved documents are relevant. Think step by step to identify important terms appearing in these documents that can be used for query expansion. Remove overly general or non-informative terms.

Original query: “climate change impact”

Top documents (snippets):

- [“Climate change impacts, risks and adaptation”,
- “Global warming likely to reach 2°C before 2050, a level associated with catastrophic impacts”,
- “Marine sediments provide clues for future impact of climate change”,

“What is climate change? A really simple guide”,
“Accounting for ocean impacts nearly doubles the social cost of carbon”]
Output: A ranked list of expansion terms and the expanded query.

” Ask ChatGPT

Ranked list of expansion terms (from pseudo-relevance feedback)

1. global warming
2. climate impacts
3. environmental impacts
4. climate risks
5. adaptation
6. ocean impacts
7. social cost of carbon
8. marine ecosystems
9. temperature increase
10. climate change effects

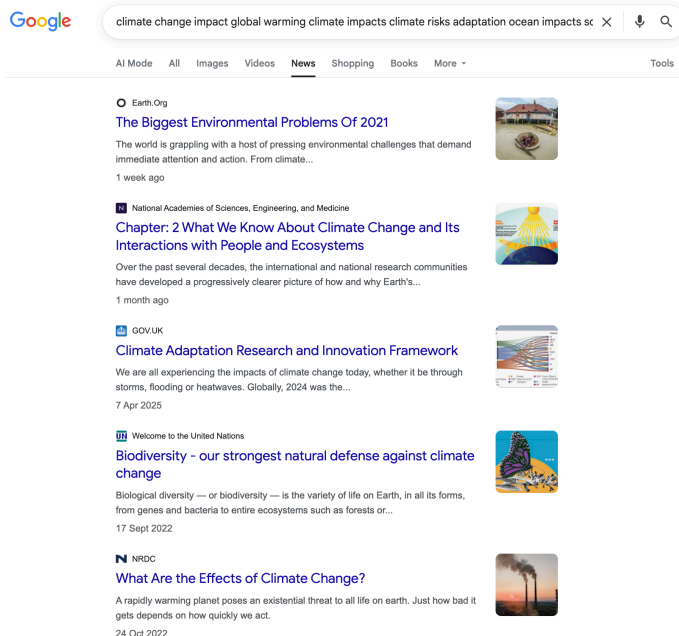
(General terms such as *guide*, *what is*, and overly generic words like *simple* are removed.)

Expanded query

climate change impact global warming climate impacts climate risks adaptation ocean impacts social cost of carbon marine ecosystems temperature increase

Search Google news again using the expanded query:

“climate change impact global warming climate impacts climate risks adaptation ocean impacts social cost of carbon marine ecosystems temperature increase”



You may discuss whether the expanded query performs better than the original query.

Question 4. Assume C is a collection of tweets. We can use the following query likelihood method to calculate the probability of a tweet d occurring for a given query Q .

$$P(Q, d) = \sum_{w \in Q, d} c(w, Q) \log \left(1 + \frac{c(w, d)}{\mu \times P(w|C)} \right) + |Q| \log \frac{\mu}{\mu + |d|}$$

Let $|d|$ and $|Q|$ are the respective lengths (size) of the tweet d and query Q , and μ is the smoothing parameter. Interpret the meaning of $c(w, d)$, $c(w, Q)$, and $P(w|C)$.

Solution:

$c(w, d)$ denotes the count of word w in the given tweet d ,

$c(w, Q)$ is the count of word w in the given query Q ,

$P(w|C)$ is the probability of the word w in the collection C that is used for model normalisation.